

## **APPENDICES**

## APPENDIX A: Evidence of Data Gathering

This presents the documentation of primary data gathering activities conducted during the problem identification phase of the study. Data collection was carried out through a structured survey questionnaire distributed to residents of various barangays within Bacoor City, Cavite from May 05, 2026 to May 12, 2026, using a mixed-mode approach consisting of online distribution via Google Forms and face-to-face administration. A total of 50 respondents from 12 barangays across Bacoor City participated in the survey, selected through *purposive-convenience sampling*, with a concentration in flood-prone barangays such as Zapote 1 and Mambog 1.

Additionally, a Key Informant Interview with the Bacoor City Disaster Risk Reduction and Management Office (BDRRMO) and the City Planning and Development Office (CPCDO) is scheduled to supplement the quantitative survey data with institutional perspectives on flood risk management in the locality.

### A.1 Data Gathering Overview

|                               |                                                                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Collection Method</b> | Online Survey Questionnaire (Google Forms) and Face-to-Face Survey Administration                                                  |
| <b>Survey Title</b>           | Flood Susceptibility and Awareness Survey                                                                                          |
| <b>Data Gathering Period</b>  | May 05, 2026 - May 12, 2026                                                                                                        |
| <b>Target Respondents</b>     | Residents of flood-prone barangays in Bacoor City, Cavite                                                                          |
| <b>Number of Respondents</b>  | <b>50 respondents (20 via Google Forms; 30 via face-to-face administration)</b>                                                    |
| <b>Survey Instrument</b>      | 13-item mixed questionnaire with Likert-scale, multiple choice, and open-ended items                                               |
| <b>Purpose</b>                | To identify flood information gaps and assess the level of flood awareness among residents of flood-prone barangays in Bacoor City |
| <b>Design Thinking Phase</b>  | Problem Identification Phase                                                                                                       |

## A.2 Respondent Profile Summary

The following tables present the distribution of respondents by barangay (Q1), length of residency (Q2) and smartphone ownership (Q3).

### Q1. Barangay of Residence

| Barangay        | No. of Respondents |
|-----------------|--------------------|
| Zapote 1        | 15                 |
| Mambog 1        | 12                 |
| Niog 3          | 5                  |
| Niog 2          | 3                  |
| Niog 1          | 2                  |
| San Nicolas 3   | 3                  |
| Molino 1        | 2                  |
| Talaba 3        | 2                  |
| Talaba 1        | 2                  |
| Talaba 2        | 1                  |
| Habay           | 1                  |
| Queens Row West | 1                  |
| Detachment      | 1                  |
| <b>TOTAL</b>    | <b>50</b>          |

### Q2. Length of Residency in Barangay

| Length of Residency in Barangay | Count     | % of Respondents |
|---------------------------------|-----------|------------------|
| Less than 1 year                | 1         | 2.0%             |
| 1–5 years                       | 5         | 10.0%            |
| 6–10 years                      | 13        | 26.0%            |
| More than 10 years              | 31        | 62.0%            |
| <b>TOTAL</b>                    | <b>50</b> |                  |

### Q3. Do you own a smartphone?

| Option       | Count     | % of Respondents |
|--------------|-----------|------------------|
| Yes          | 49        | 98.0%            |
| No           | 1         | 2.0%             |
| <b>TOTAL</b> | <b>50</b> |                  |

### A.3 Survey Instrument

The survey questionnaire consisted of 13 items organized into four thematic areas: (1) Respondent Profile, (2) Flood Experience and Hazard Exposure, (3) Flood Information Assessment, and (4) Decision-Making and Suggestions. The instrument was distributed online via Google Forms and a face-to-face administration.

### Survey Questionnaire Structure

| Part                              | Items   | Coverage                                                                                             |
|-----------------------------------|---------|------------------------------------------------------------------------------------------------------|
| Part I : Respondent Profile       | Q1–Q3   | Barangay, length of residency, smartphone ownership                                                  |
| Part II : Flood Experience        | Q4–Q8   | Flood frequency, severity, warning sources, flood-area awareness, effects of flooding                |
| Part III : Information Assessment | Q9–Q11  | Satisfaction with flood information, timeliness of warnings, clarity of communication (Likert)       |
| Part IV : Decision-Making         | Q12–Q13 | Biggest problem in getting flood information, wrong decisions due to lack of flood info (open-ended) |

## APPENDIX B: Survey Results

This appendix presents the complete quantitative and qualitative findings of the Flood Susceptibility and Awareness survey administered to 50 residents of flood-prone barangays in Bacoor City, Cavite. The survey was conducted between May 05, 2026 to May 12, 2026, using a mixed-mode approach consisting of online distribution via Google Forms and face-to-face administration. Results are organized thematically according to the survey's four-part structure.

## B.1 Flood Experience and Hazard Exposure

### Q4. How frequently does your area experience flooding?

| Option                                      | Count     | % of Respondents |
|---------------------------------------------|-----------|------------------|
| Never                                       | 1         | 2.0%             |
| Rarely (once every few years)               | 3         | 6.0%             |
| Occasionally (once or twice a year)         | 2         | 4.0%             |
| Frequently (several times a year)           | 17        | 34.0%            |
| Very frequently (almost every rainy season) | 27        | 54.0%            |
| <b>TOTAL</b>                                | <b>50</b> |                  |

**Key Finding: 92.0% of respondents (Occasionally + Frequently + Very Frequently) experience flooding in their area, confirming widespread flood exposure in Bacoar City.**

### Q5. Typical severity of flooding in your area?

| Option                        | Count     | % of Respondents |
|-------------------------------|-----------|------------------|
| No flooding                   | 1         | 2.0%             |
| Minor (ankle-deep)            | 8         | 16.0%            |
| Moderate (knee-deep)          | 12        | 24.0%            |
| Severe (waist-deep or higher) | 29        | 58.0%            |
| <b>TOTAL</b>                  | <b>50</b> |                  |

**Key Finding: 82.0% of respondents reported Moderate to Severe flooding (knee-deep or higher), indicating high flood risk severity among the surveyed population.**

### Q6. How do you receive flood warnings? (Select all that apply — % is of total respondents)

| Option             | Count | % of Respondents |
|--------------------|-------|------------------|
| Text/SMS alerts    | 32    | 64.0%            |
| Social media       | 33    | 66.0%            |
| Barangay PA system | 15    | 30.0%            |

|                                   |            |       |
|-----------------------------------|------------|-------|
| News/TV                           | 21         | 42.0% |
| Neighbors/word of mouth           | 22         | 44.0% |
| I receive no warning              | 2          | 4.0%  |
| <b>TOTAL (multiple responses)</b> | <b>125</b> |       |

**Key Finding: Social media (66%) and SMS alerts (64%) are the dominant warning channels among respondents, while only 30% receive warnings through the official Barangay PA system. Notably, 4% report receiving no warning at all, indicating over-reliance on informal channels and gaps in official flood communication reach.**

**Q7. How aware are you of flood-prone areas in Bacoar?**

| Option           | Count     | % of Respondents |
|------------------|-----------|------------------|
| Well-informed    | 10        | 20.0%            |
| Somewhat aware   | 30        | 60.0%            |
| Not sure         | 4         | 8.0%             |
| Not aware at all | 6         | 12.0%            |
| <b>TOTAL</b>     | <b>50</b> |                  |

**Key Finding: Only 20% of respondents consider themselves well-informed about flood-prone areas in Bacoar City, while 80% fall under "Somewhat aware," "Not sure," or "Not aware at all" — highlighting a significant gap in flood risk awareness among the surveyed population.**

**Q8. What effects has flooding had on you? (Select all that apply — % is of total respondents)**

| Option                                        | Count      | % of Respondents |
|-----------------------------------------------|------------|------------------|
| Property damage                               | 39         | 78.0%            |
| Disruption to daily activities (work, school) | 48         | 96.0%            |
| Health risks                                  | 42         | 84.0%            |
| Displacement or evacuation                    | 19         | 38.0%            |
| Loss of livelihood                            | 18         | 36.0%            |
| None of the above                             | 0          | 0.0%             |
| <b>TOTAL (multiple responses)</b>             | <b>166</b> |                  |

**Key Finding: 96.0% of respondents reported disruption to daily activities, and 84.0% reported health risks — confirming that flooding has broad and severe consequences on daily life.**

#### B.4 Flood Information Assessment (Likert Scale Summary)

The following items (Q9, Q10, Q11) used a 1–5 Likert scale (1 = lowest, 5 = highest) to measure respondents' perceptions of flood information quality.

*The following scale was used to interpret mean scores:*

| Mean Score Range | Interpretation            |
|------------------|---------------------------|
| 1. 00 - 1.80     | Very Dissatisfied / Never |
| 1.81 - 2.60      | Dissatisfied / Rarely     |
| 2.61 - 3.40      | Neutral / Sometimes       |
| 3.41 – 4.20      | Satisfied / Usually       |
| 4.21 – 5.00      | Very Satisfied / Always   |

| Question                                 | Rating 1 | Rating 2 | Rating 3 | Rating 4 | Rating 5 | Mean Score  | Interpretation      |
|------------------------------------------|----------|----------|----------|----------|----------|-------------|---------------------|
| Q9. Satisfaction with flood information  | 13       | 9        | 16       | 8        | 4        | <b>2.62</b> | Neutral / Sometimes |
| Q10. Timeliness of flood warnings        | 8        | 12       | 19       | 9        | 2        | <b>2.70</b> | Neutral / Sometimes |
| Q11. Clarity of flood risk communication | 9        | 10       | 20       | 8        | 3        | <b>2.72</b> | Neutral / Sometimes |

**Key Finding: All three Likert items (Q9, Q10, Q11) yielded mean scores of 2.62–2.72 (Neutral/Sometimes), indicating that respondents are generally dissatisfied with the current quality, timeliness, and clarity of flood information. This indicates that respondents are generally dissatisfied with the current, timeliness, and clarity of flood information they receive.**

## B.5 Decision-Making and Open-Ended Responses

### Q12. Biggest Problem in Getting Flood Information (Open-Ended)

35 substantive responses analyzed; 15 respondents answered 'n/a' or left blank. Themes are not mutually exclusive — some responses reflect more than one theme.

| Theme                                      | Frequency | % (n=50) | Description                                                                                                                 |
|--------------------------------------------|-----------|----------|-----------------------------------------------------------------------------------------------------------------------------|
| Late / Delayed Information                 | 13        | 26%      | Warnings and advisories arrive too late, after flooding has already begun or worsened.                                      |
| Lack of Flood Information / No Warnings    | 10        | 20%      | Residents receive little to no flood-related information from LGU or barangay officials.                                    |
| Unclear or Inaccurate Information          | 6         | 12%      | Information provided is vague, confusing, or factually incorrect — making it difficult to assess actual risk.               |
| Sudden / Unpredictable Flooding            | 4         | 8%       | Flooding occurs rapidly and without prior signs, leaving no time to prepare even when residents are attentive.              |
| Loss of Connectivity / Power Outages       | 3         | 6%       | Internet and power loss during typhoons disrupts access to real-time flood information.                                     |
| Lack of Location-Specific Risk Information | 3         | 6%       | Residents cannot determine whether their specific street or area is at risk — information is too general.                   |
| Technology / Accessibility Barriers        | 2         | 4%       | Not all residents — particularly senior citizens — have smartphones or internet access to receive digital alerts.           |
| No Problem / Already Accustomed            | 5         | 10%      | Respondents expressed no concern, citing habituation to flooding or satisfaction with existing channels (TV, social media). |
| No Response (N/A)                          | 15        | 30%      | Respondent left the question blank or answered 'n/a'.                                                                       |

**Key Finding: Late/Delayed Information (26%) and Lack of Flood Information/No Warnings (20%) are the two most common themes, together accounting for 38% of all responses — confirming a significant gap in timeliness and availability of flood communication in Bacoor City.**

## **Q12. Notable Quotes per Theme**

### **Late / Delayed Information**

- *"laging huli ang balita"*
- *"Timeliness - usually, late notice or lack of notice or information"*
- *"Last minute advisories."*
- *"siguro mabagal ang pag update ng mga news kung kelan baha na agad"*

### **Lack of Flood Information / No Warnings**

- *"No immediate and effective information from the LGU. no actions too"*
- *"No text message from the barangay when the flood is coming"*
- *"lack of announcement sometimes, specially at nighttime"*
- *"Lack of information from the City Government"*

### **Unclear or Inaccurate Information**

- *"minsan mali yung impormasyon na binigay"*
- *"Sometimes the information comes unclear"*
- *"di nila sinasabi hanggang saan yung baha"*
- *"no specific information, basta umulan expected na babaha na"*

### **Sudden / Unpredictable Flooding**

- *"sobrang rare, magugulat ka nalang na baha na agad isang minuto lang"*
- *"di kami naiinformed kaagad"*

### **Loss of Connectivity / Power Outages**

- *"nawawalan ng kuryente at internet pag may bagyo minsan"*

### **Lack of Location-Specific Risk Information**

- *"maganda kung alam mo kung atleast saan ang mga flood prone areas depende sa lakas at tagal ng ulan"*
- *"kulang yong mga infos at siguro walang smart planning na pwede gawin base sa lugar"*
- *"no proper drainage system and not enough flood information that could help or educate people"*

### **Technology / Accessibility Barriers**

- *"most of the residents here are Senior Citizens which are not fond of using smartphones. They somewhat rely on radios or televisions"*

### **No Problem / Already Accustomed**

- *"hindi naman problema ang flood information dahil may tv news at social media at sanay na kami"*
- *"wala naman kasi sanay na din at alam na din bahain dito sa tagal ko ng nakatira dito"*

**Q13: Have you ever made a wrong decision due to lack of flood information?**

| Option                      | Count     | % of Respondents |
|-----------------------------|-----------|------------------|
| Yes — made a wrong decision | 32        | 64.0%            |
| No                          | 18        | 36.0%            |
| <b>TOTAL</b>                | <b>50</b> |                  |

**Q13.a: Thematic Analysis (from 32 Yes respondents)**

| Theme                                  | Frequency | % of Yes (n=32) | Description                                                                                                           |
|----------------------------------------|-----------|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| Took a Flooded Road / Route            | 11        | 34%             | Respondents passed through flooded streets unaware of water depth, leading to dangerous situations or vehicle damage. |
| Stranded / Unable to Travel            | 8         | 25%             | Respondents left home or workplace without knowing flooding would block routes, resulting in being stranded.          |
| Stayed Home / Did Not Evacuate in Time | 6         | 19%             | Respondents remained home assuming flooding would be minor, only to find water levels rising unexpectedly.            |
| Misjudged Flood Depth                  | 4         | 13%             | Respondents relied on secondhand or general information that underestimated actual water depth.                       |
| Wrong Evacuation Decision              | 3         | 9%              | Respondents evacuated to an area that turned out to be more flooded than their original location.                     |

**Key Finding: 64% of respondents (32 of 50) have made a wrong decision due to insufficient flood information, providing strong empirical support for the study's problem statement.**

**Q13 — Notable Quotes per Theme****Took a Flooded Road / Route**

- "took a flooded road going to palengke and work"
- "nagsise ako dumaan kami sa niog1, malapit sa cvsu"
- "motorcycle got drowned in the flood"

**Stranded / Unable to Travel**

- "pumasok sa school, na stranded"

- *"Since flooded road is very common in our municipality, I experienced a 3-hour traffic jam"*
- *"Stayed at work then when we were finally allowed to go home, roads were already flooded"*

#### **Stayed Home / Did Not Evacuate in Time**

- *"stayed at home kahit may signal no.3 warning"*
- *"I made the wrong call to stay home — the only info I had was a late announcement"*
- *"kala namin malakas lang ulan, pero chineck ng misis ko at pinasok na kami ng tubig"*

#### **Wrong Evacuation Decision**

- *"We evacuated to Talaba but the flood was higher there — almost shoulder level"*
- *"hindi nag angat kaagad ng gamit"*

#### **Misjudged Flood Depth**

- *"the neighbor told me it was ankle deep but it was deeper than that"*
- *"kalako hanggang bewang lang ang baha, yun pala lagpas tao"*

#### **Analyst Notes**

1. Themes were derived through manual thematic analysis of all 50 responses. N/A and blank answers were excluded from thematic coding but included in frequency counts.
2. Q12 findings confirm a significant gap in timeliness and clarity of flood information — the two most recurring themes together account for 38% of all responses.
3. Q13 findings indicate that 64% of respondents (32 of 50) have made a wrong decision due to insufficient flood information, providing strong empirical support for the study's problem statement.
4. Overlapping themes (e.g., a response citing both late warnings and unclear information) were tagged under the primary theme based on the respondent's emphasis.

#### **Proofs of data gathering:**

Gdrive:  Flood Susceptibility

Spreadsheet:

 Flood Susceptibility - SurveyTally